Vector-Valued Functions

Position, velocity, and acceleration vectors

1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

2 For the same particle, find the speed at $t = 2$.

Integrating vector-valued functions

3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

Total distance traveled

4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

Vectors and parametric motion combined

5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.

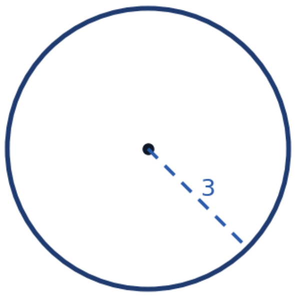
Unit tangent vectors

7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.

Motion on a circle and centripetal acceleration

9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).



10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.

Displacement vs. distance for vector motion

11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

Vector operations review

13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.

14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

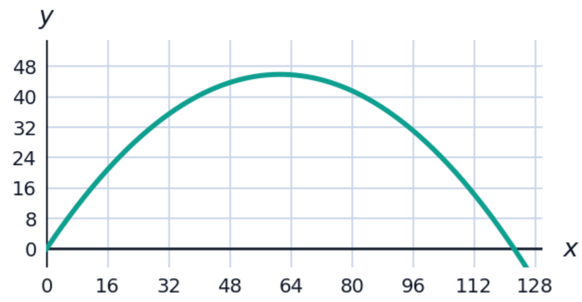
Curvature-related speed problems

15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

Projectile motion as a vector-valued function

17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).



18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

Finding when a particle is at rest

19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

20 For the same particle, find the times when only the horizontal velocity component is 0.

Vector-valued functions and arc length

21 State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

22 Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.